

Chapter Two

Solution of First Order Differential Equation

Methods of Solution of Differential Equations: There are various methods to solve a first order differential equation depending on the characteristics of the equation. In this chapter, we will learn the following methods and solve some related problems:

- Variables Separable Equation
- Homogeneous Differential Equation
- Equations Reducible to Homogeneous Differential Equations
- Exact Differential Equations
- Special Types of Integrating factor
- Integrating factor by Inspection
- Linear Differential Equations
- Bernoulli's Differential Equations
- Ricatti's Differential Equations

Section-2.1

Variables Separable Equation

2.1.1 First Order Differential Equation

A differential equation of the form $M(x, y)dx + N(x, y)dy = 0$... (1)

where M and N both are functions of x and y , is called first order differential equation.

Example: The equation $y\sqrt{x^2-1}dx + x\sqrt{y^2-1}dy = 0$ is a first order differential equation of the form (1).

2.1.2 Separation of Variables

Separable Equation: Equation (1) can be represented of the form

$$F(x)G(y)dx + f(x)g(y)dy = 0 \quad \dots (2)$$

then the equation is called an equation with variables separable or simply a separable equation.

Rearranging the equation (2), we obtain

$$\frac{F(x)}{f(x)}dx + \frac{g(y)}{G(y)}dy = 0$$

$$\text{or, } A(x)dx + B(y)dy = 0 \quad \dots (3)$$

The left member of (3) can be written as total differential of F , that is $dF = A(x)dx + B(y)dy$. Then the desired solution of the equation is $F(x, y) = C$, where C is a arbitrary constant.

2.1.3 Equation Reducible to Variables Separation

If the differential equation is of the form $\frac{dy}{dx} = f(ax + by + c)$, then it can be reduced to an equation in which variables can be separated by the method of substitution. i.e., using substitution $ax + by + c = z$.

Given differential equation: $\frac{dy}{dx} = f(ax + by + c) \dots (4)$

We put $ax + by + c = z$ then $a + b\frac{dy}{dx} = \frac{dz}{dx}$

$$\begin{aligned} \Rightarrow b\frac{dy}{dx} &= \frac{dz}{dx} - a \\ \Rightarrow \frac{dy}{dx} &= \frac{1}{b}\left(\frac{dz}{dx} - a\right) \dots (5) \end{aligned}$$

From (4) and (5), we obtain

$$\Rightarrow \frac{1}{b}\left(\frac{dz}{dx} - a\right) = f(z)$$

$$\Rightarrow \frac{dz}{dx} = a + bf(z)$$

$$\Rightarrow \frac{dz}{a + bf(z)} = dx$$

The above equation is in the form of variables separated. Now, the solution will obtained by integrating.

Solved Problems:

Example-1 Solve the equation: $\frac{dy}{dx} + \sqrt{\frac{1-y^2}{1-x^2}} = 0$

Solution Given differential equation

$$\begin{aligned} \frac{dy}{dx} + \sqrt{\frac{1-y^2}{1-x^2}} &= 0 \\ \Rightarrow \frac{dx}{\sqrt{1-x^2}} + \frac{dy}{\sqrt{1-y^2}} &= 0 \end{aligned}$$

$$\Rightarrow \int \frac{dx}{\sqrt{1-x^2}} + \int \frac{dy}{\sqrt{1-y^2}} = C \quad ; [\text{Integrating}]$$

$$\therefore \sin^{-1}x + \sin^{-1}y = C$$

Which is the desired solution.

Example-2 Solve the equation: $x\sqrt{1-y^2}dx + y\sqrt{1-x^2}dy = 0$

Solution Given differential equation,

$$\begin{aligned} x\sqrt{1-y^2}dx + y\sqrt{1-x^2}dy &= 0 \\ \Rightarrow \frac{x}{\sqrt{1-x^2}}dx + \frac{y}{\sqrt{1-y^2}}dy &= 0 \\ \Rightarrow \frac{-2x}{\sqrt{1-x^2}}dx + \frac{-2y}{\sqrt{1-y^2}}dy &= 0 \end{aligned}$$

$$\Rightarrow \int \frac{d(1-x^2)}{\sqrt{1-x^2}} + \int \frac{d(1-y^2)}{\sqrt{1-y^2}} = 0 \quad ; [\text{Integrating}]$$

$$\Rightarrow \frac{(1-x^2)^{-1/2+1}}{-1/2+1} + \frac{(1-y^2)^{-1/2+1}}{-1/2+1} = 2c$$

$$\therefore \sqrt{1-x^2} + \sqrt{1-y^2} = c$$

This is the required solution.

Example-3 Solve the equation: $y\sqrt{x^2-1}dx + x\sqrt{y^2-1}dy = 0$

Solution Given differential equation

$$y\sqrt{x^2-1}dx + x\sqrt{y^2-1}dy = 0$$

$$\Rightarrow \frac{\sqrt{x^2-1}}{x}dx + \frac{\sqrt{y^2-1}}{y}dy = 0$$

$$\Rightarrow \frac{x^2-1}{x\sqrt{x^2-1}}dx + \frac{y^2-1}{y\sqrt{y^2-1}}dy = 0$$

$$\Rightarrow \frac{x}{\sqrt{x^2-1}}dx - \frac{dx}{x\sqrt{x^2-1}} - \frac{dy}{y\sqrt{y^2-1}} + \frac{y}{\sqrt{y^2-1}}dy = 0$$

$$\Rightarrow \frac{\frac{1}{2}d(x^2-1)}{\sqrt{x^2-1}} + \frac{\frac{1}{2}d(y^2-1)}{\sqrt{y^2-1}} = \frac{dx}{x\sqrt{x^2-1}} + \frac{dy}{y\sqrt{y^2-1}}$$

Integrating on both sides, we get

$$\Rightarrow \frac{1}{2} \frac{(x^2-1)^{-1/2+1}}{-1/2+1} + \frac{1}{2} \frac{(y^2-1)^{-1/2+1}}{-1/2+1} = \sec^{-1}x + \sec^{-1}y + c$$

$$\therefore \sqrt{x^2-1} + \sqrt{y^2-1} = \sec^{-1}x + \sec^{-1}y + c$$

This is the desired solution.

Example-4 Solve the equation: $ydx + (1+x^2)\tan^{-1}x dy = 0$

Solution Given differential equation,

$$ydx + (1+x^2)\tan^{-1}x dy = 0$$

$$\Rightarrow \frac{dx}{(1+x^2)\tan^{-1}x} + \frac{1}{y}dy = 0$$

$$\Rightarrow \frac{d(\tan^{-1}x)}{\tan^{-1}x} + \frac{dy}{y} = 0$$

$$\Rightarrow \int \frac{d(\tan^{-1}x)}{\tan^{-1}x} + \int \frac{dy}{y} = c$$

$$\Rightarrow \ln \tan^{-1}x + \ln y = \ln c$$

$$\Rightarrow \ln(y \tan^{-1}x) = \ln c$$

$$\therefore y \tan^{-1}x = c$$

This is the desired solution.

Example-5 Solve the equation: $\sec^2 x \tan y dx + \sec^2 y \tan x dy = 0$

Example-6 Solve the equation: $\frac{dy}{dx} + \frac{y^2+y+1}{x^2+x+1} = 0$

Example-7 Solve the following differential equations:

$$(i) \ln\left(\frac{dy}{dx}\right) = ax + by \quad (ii) \ln\left(\frac{dy}{dx}\right) = 3x + 4y$$

Example-8 Solve the equation: $x(e^y + 4)dx + e^{x+y}dy = 0$.

Example-9 Solve the equation: $(e^x + 1)ydy = (y+1)e^x dx$.

Example-10 Solve the equation: $\frac{dy}{dx} + 1 = e^{x-y}$.

Example-11 Solve the equation: $x\sqrt{y}dx + (1+y)\sqrt{1+x}dy = 0$.

Example-12 Solve the equation: $y - x\frac{dy}{dx} = a\left(y^2 + \frac{dy}{dx}\right)$

Example-13 Solve the equation: $y(1+xy)dx + x(1-xy)dy = 0$

Solution Given differential equation,

$$y(1+xy)dx + x(1-xy)dy = 0$$

$$\Rightarrow ydx + xdy + xy(ydx - xdy) = 0$$

$$\Rightarrow d(xy) + x^2y^2\left(\frac{dx}{x} - \frac{dy}{y}\right) = 0$$

$$\Rightarrow \frac{d(xy)}{(xy)^2} + \left(\frac{dx}{x} - \frac{dy}{y}\right) = 0$$

Integrating on both sides, we get

$$\Rightarrow -\frac{1}{xy} + \ln x - \ln y = c$$

$$\therefore \ln \frac{x}{y} - \frac{1}{xy} = c$$

which is the desired solution.

Example-14 Solve the equation: $\frac{dy}{dx} = \sqrt{y-x}$

Solution Given differential equation

$\frac{dy}{dx} = \sqrt{y-x}$ $\Rightarrow 2v \frac{dv}{dx} + 1 = v$ $\Rightarrow 2v \frac{dv}{dx} = v - 1$ $\Rightarrow \frac{2v}{v-1} dv = dx$ $\Rightarrow \int \frac{2v}{v-1} dv = \int dx$ $\Rightarrow 2 \int \left(1 + \frac{1}{v-1}\right) dv = \int dx$ $\Rightarrow 2v + 2 \ln(v-1) = x + C$ $\therefore 2\sqrt{y-x} + 2 \ln(\sqrt{y-x} - 1) = x + C$	Put $y-x = v^2$ $\Rightarrow y = v^2 + x$ $\therefore \frac{dy}{dx} = 2v \frac{dv}{dx} + 1$
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This is the required solution.

Example-15 Solve the following differential equations:

(i) $\sin^{-1}\left(\frac{dy}{dx}\right) = x + y$ (ii) $\cos^{-1}\left(\frac{dy}{dx}\right) = x + y$ (iii) $\frac{dy}{dx} = \sin(x+y) + \cos(x+y)$

Solution (iii) Given differential equation

$\frac{dy}{dx} = \sin(x+y) + \cos(x+y)$ $\Rightarrow \frac{dv}{dx} - 1 = \sin v + \cos v$ $\Rightarrow \frac{dv}{dx} = 1 + \cos v + \sin v = 2 \cos^2 \frac{v}{2} + 2 \sin \frac{v}{2} \cos \frac{v}{2}$ $\Rightarrow \frac{dv}{dx} = 2 \cos \frac{v}{2} \left(\cos \frac{v}{2} + \sin \frac{v}{2} \right) = 2 \cos^2 \frac{v}{2} \left(1 + \tan \frac{v}{2} \right)$	$\Rightarrow \frac{dv}{2 \cos^2 \frac{v}{2} \left(1 + \tan \frac{v}{2} \right)} = dx$ $\Rightarrow \int \frac{\frac{1}{2} \sec^2 \frac{v}{2}}{1 + \tan \frac{v}{2}} dv = \int dx \quad ; \text{ [Integrating]}$ $\Rightarrow \ln \left(1 + \tan \frac{v}{2} \right) = x + c$ $\therefore \ln \left(1 + \tan \frac{x+y}{2} \right) = x + c \quad [\because v = x + y]$
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This is the desired solution.

Example-17 Solve the equation: $(x+y)^2 \frac{dy}{dx} = a^2$

Example-18 Solve the equation: $\frac{dy}{dx} = (2x+3y-5)^2$

Example-19 Solve the equation: $(x+y+1) \frac{dy}{dx} = 1$

Example-20 Solve the equation: $(x+y+1)dx - (2x+2y+1)dy = 0$

Section 2.2

Homogeneous Differential Equations

2.2.1 Homogeneous Equations

Polynomial in which all terms are of the same degree, such as

$$x^3 + 3x^2y - xy^2 + y^3, \quad x^3 - y^3, \quad x^4y - 5x^2y^3 + xy^4$$

are called homogeneous polynomials.

Definition: An equation in which sum of the powers of the variables in every terms are equal, then the equation is called homogeneous equation.

Examples: $x^3 + 3x^2y - xy^2 + y^3 = 0$, $x^3 - y^3 = 0$, $x^4y - 5x^2y^3 + xy^4 = 0$ are all homogeneous equations.

Or, Let us consider an equation $f(x, y) = 0$. If the function $f(x, y)$ can be expressed as $f(\lambda x, \lambda y) = \lambda^n f(x, y)$, then the equation is called homogeneous equation of x and y of the degree n .

Examples: $f(x, y) = \frac{x^2 + 3xy}{x - 2y}$ is a homogeneous function of degree one in x and y , because the function can be expressed as $f(\lambda x, \lambda y) = \lambda^1 f(x, y)$.

2.2.2 Homogeneous Differential Equations

Let us consider the differential equation of the form

$$M(x, y)dx + N(x, y)dy = 0 \quad \dots (1)$$

$$\text{or } \frac{dy}{dx} = \frac{M(x, y)}{N(x, y)} \quad \dots (2)$$

If the functions $M(x, y)$ and $N(x, y)$ in equations (1) and (2), are homogeneous functions of x and y of the same degree, then the equations are called homogeneous differential equations.

Examples: $(x^2 + y^2)dx - 2xydy = 0$

$$\text{and } \frac{dy}{dx} = \frac{y}{x} + \tan \frac{y}{x}$$

both are all homogeneous differential equations.

Or, If the function $f(x, y)$ in the differential equation $\frac{dy}{dx} = f(x, y)$ can be expressed as a function of the ratio $\frac{y}{x}$ only, then the equation is said to be homogeneous. Such equations can always be transformed into separable equations by change of the dependent variable.

Example: $\frac{dy}{dx} = \frac{y - 4x}{x - y}$ is a homogeneous differential equation, because the right side of the differential equation can

be expressed as a function of the ratio $\frac{y}{x}$ only.

$$\text{i.e., } \frac{dy}{dx} = \frac{(y/x) - 4}{1 - (y/x)}$$

Solved Problems:

Example-1 Solve the differential equation $\frac{dy}{dx} = \frac{y - 4x}{x - y}$

Solution Given differential equation,

$$\frac{dy}{dx} = \frac{y - 4x}{x - y} \quad \dots (1)$$

Dividing both the numerator and denominator of the right side of the equation (1) by x , we obtain

$$\frac{dy}{dx} = \frac{(y/x) - 4}{1 - (y/x)} \quad \dots (2)$$

Right side of the equation (2) is a function of the ratio $\frac{y}{x}$ only, so the differential equation (1) or (2) is a homogeneous differential equation.

Now, we introduce a new dependent variable v so that $v = \frac{y}{x}$, or $y = xv(x)$. Then $\frac{dy}{dx}$ can be expressed in terms of x, v and $\frac{dv}{dx}$. That is, we get

$$\frac{dy}{dx} = v + x \frac{dv}{dx} \quad \dots (3)$$

Eliminating $\frac{dv}{dx}$ from (2) and (3) and replacing $\frac{y}{x}$ by v , we obtain

$$\begin{aligned} v + x \frac{dv}{dx} &= \frac{v-4}{1-v} \\ \Rightarrow x \frac{dv}{dx} &= \frac{v-4}{1-v} - v \\ \Rightarrow x \frac{dv}{dx} &= \frac{v^2-4}{1-v} \quad \dots (4) \end{aligned}$$

We observe that, the equation (4) is separable. Now, equation (4) can be written as

$$\begin{aligned} \left(\frac{1-v}{v^2-4} \right) dv &= \frac{dx}{x} \\ \Rightarrow \frac{dv}{v^2-2^2} - \frac{v dv}{v^2-4} &= \frac{dx}{x} \end{aligned}$$

Integrating on both sides we obtain the solution of the equation (4), v implicitly in terms of x .

$$\frac{1}{2 \cdot 2} \ln \left| \frac{v-2}{v+2} \right| - \frac{1}{2} \ln |v^2-4| = \ln |x| + \frac{1}{4} \ln C$$

where $\frac{1}{4} \ln C$ integrating constant.

Replacing v by $\frac{y}{x}$, we obtain

$$\begin{aligned} \ln \left| \frac{\frac{y}{x}-2}{\frac{y}{x}+2} \right| - 2 \ln \left| \frac{y^2}{x^2} - 4 \right| &= 2 \ln |x| + \ln C \\ \Rightarrow \ln \left| \frac{y-2x}{y+2x} \right| - 2 \ln \left| \frac{y^2-4x^2}{x^2} \right| &= 2 \ln |x| + \ln C \\ \Rightarrow \ln \left| \left(\frac{y-2x}{y+2x} \right) \frac{x^4}{(y^2-4x^2)^2} \right| &= \ln |x^2 C| \\ \Rightarrow \frac{x^4}{(y+2x)^2 (y^2-4x^2)} &= x^2 C \\ \therefore x^2 &= C(y+2x)^2 (y^2-4x^2) \end{aligned}$$

which is the required solution of the given differential equation.

Example-2 Solve the Differential Equation $(x^2 + y^2)dx + xy dy = 0$

Solution Given differential equation,

$$(x^2 + y^2)dx + xy dy = 0$$

Which is a homogeneous differential equation.

Let us put $y = vx$ or, $dy = v dx + x dv$, then equation (1) becomes

$$\begin{aligned} (x^2 + v^2 x^2)dx + vx^2(v dx + x dv) &= 0 \\ \Rightarrow (1 + v^2)dx + v(v dx + x dv) &= 0 \\ \Rightarrow (1 + v^2 + v^2)dx + vx dv &= 0 \\ \Rightarrow (1 + 2v^2)dx + vx dv &= 0 \\ \Rightarrow \frac{dx}{x} + \frac{v}{1 + 2v^2} dv &= 0 \end{aligned}$$

$$\Rightarrow \int \frac{dx}{x} + \frac{1}{4} \int \frac{4v}{2v^2 + 1} dv = 0 \quad [\text{Integrating}]$$

$$\Rightarrow \ln|x| + \frac{1}{4} \ln|2v^2 + 1| = \ln|C|$$

$$\Rightarrow 4\ln|x| + \ln|2v^2 + 1| = 4\ln|C|$$

$$\Rightarrow x^4(2v^2 + 1) = C^4$$

$$\Rightarrow x^4 \left(2 \cdot \frac{y^2}{x^2} + 1 \right) = C^4$$

$$\therefore x^2(x^2 + 2y^2) = C^4$$

Which is the required solution of the given differential equation.

Example-3 Solve the equation: $(x^2 + y^2)dx - 2xy dy = 0$

Solution Given differential equation,

$$(x^2 + y^2)dx - 2xy dy = 0 \quad \dots (1)$$

which is a homogeneous differential equation.

We put $y = vx$ then $dy = v dx + x dv$... (2)

Substituting these values in equation (1), we get

$$\begin{aligned} (x^2 + v^2 x^2)dx - 2vx^2(v dx + x dv) &= 0 \\ \Rightarrow (1 + v^2)dx - 2v(v dx + x dv) &= 0 \\ \Rightarrow (1 + v^2 - 2v^2)dx - 2vx dv &= 0 \\ \Rightarrow (1 - v^2)dx - 2vx dv &= 0 \end{aligned}$$

$$\Rightarrow \frac{dx}{x} - \frac{2v}{1 - v^2} dv = 0$$

$$\Rightarrow \frac{2v}{v^2 - 1} dv + \frac{dx}{x} = 0$$

Integrating on both sides, we get

$$\int \frac{2v dv}{v^2 - 1} + \int \frac{dx}{x} = 0$$

$$\Rightarrow \ln(v^2 - 1) + \ln x = \ln c$$

$$\Rightarrow \ln(v^2 - 1)x = \ln c$$

$$\Rightarrow \left(\frac{y^2}{x^2} - 1 \right) x = c$$

$$\Rightarrow \frac{y^2 - x^2}{x} = c$$

$\therefore y^2 - x^2 = cx$
which is the desired solution.

Example-4 Solve the equation: $(x + y)dy + (x - y)dx = 0$

Example-5 Solve the equation: $(x^2 - xy + y^2)dx = xy dy$

Example-6 Solve the equation: $xdy - ydx = \sqrt{x^2 + y^2} dx$

Example-7 Solve the equation: $2xdy - 2ydx - \sqrt{x^2 + 4y^2} dx = 0$

Solution Given differential equation,

$$2xdy - 2ydx - \sqrt{x^2 + 4y^2} dx = 0 \dots (1)$$

Which is a homogeneous differential equation.

We put $y = vx$, then $dy = v dx + x dv$

Substituting these values in equation (1), we get

$$\begin{aligned} \Rightarrow 2vx dx + 2x^2 dv - 2vx dx - x\sqrt{1 + 4v^2} dx &= 0 \\ \Rightarrow 2x^2 dv - x\sqrt{1 + 4v^2} dx &= 0 \\ \Rightarrow \frac{2dv}{\sqrt{1 + 4v^2}} &= \frac{dx}{x} \end{aligned}$$

Integrating on both sides, we get

$$\int \frac{2dv}{\sqrt{1 + 4v^2}} = \int \frac{dx}{x}$$

$$\Rightarrow \int \frac{2dv}{2\sqrt{(1/2)^2 + v^2}} = \int \frac{dx}{x}$$

$$\Rightarrow \ln \left(v + \sqrt{\left(\frac{1}{2} \right)^2 + v^2} \right) = \ln x + \ln c$$

$$\Rightarrow \ln \left(v + \sqrt{\frac{1}{4} + v^2} \right) = \ln cx$$

$$\Rightarrow \frac{y}{x} + \sqrt{\frac{1}{4} + \frac{y^2}{x^2}} = cx$$

$$\Rightarrow \frac{y}{x} + \sqrt{\frac{x^2 + 4y^2}{4x^2}} = cx$$

$$\Rightarrow \frac{y}{x} + \frac{\sqrt{x^2 + 4y^2}}{2x} = cx$$

$\therefore 2y + \sqrt{x^2 + 4y^2} = 2cx^2$
which is the required solution.

Example-8 Solve the equation: $(x^3 + y^2\sqrt{x^2 + y^2})dx - xy\sqrt{x^2 + y^2}dy = 0$

Example-9 Solve the equation: $\frac{dy}{dx} = \frac{y}{x} + \tan \frac{y}{x}$

Example-10 Solve the equation: $\frac{dy}{dx} = \frac{y}{x} + \sin \frac{y}{x}$

Example-11 Solve the equation: $\left(x \sin \frac{y}{x} - y \cos \frac{y}{x}\right)dx + x \cos \frac{y}{x} dy = 0$

Solution Given differential equation,

$$\begin{aligned} &\left(x \sin \frac{y}{x} - y \cos \frac{y}{x}\right)dx + x \cos \frac{y}{x} dy = 0 \\ \Rightarrow &x \cos \frac{y}{x} dy = -\left(x \sin \frac{y}{x} - y \cos \frac{y}{x}\right)dx \\ \Rightarrow &\frac{dy}{dx} = \frac{y \cos(y/x) - x \sin(y/x)}{x \cos(y/x)} \end{aligned}$$

Which is a homogeneous differential equation.

We put $y = vx$ then $dy = v dx + x dv$

Substituting these values of in equation (1), we get

$$\begin{aligned} &(x \sin v - vx \cos v)dx + x \cos v(vdx + xdv) = 0 \\ \Rightarrow &(\sin v - v \cos v)dx + \cos v(vdx + xdv) = 0 \end{aligned}$$

$$\Rightarrow \sin v dx - v \cos v dx + v \cos v dx + x \cos v dv = 0$$

$$\Rightarrow \sin v dx + x \cos v dv = 0$$

$$\Rightarrow \cot v dv + \frac{dx}{x} = 0$$

$$\Rightarrow \int \cot v dv + \int \frac{dx}{x} = 0 \text{ [Integrating]}$$

$$\Rightarrow \ln(\sin v) + \ln x = \ln c$$

$$\Rightarrow \ln(x \sin v) = \ln c$$

$$\therefore x \sin\left(\frac{y}{x}\right) = c$$

which is the required solution.

Example-12 Solve the equation: $ye^{x/y}dx + (y - xe^{x/y})dy = 0$

Example-13 Solve the equation: $(1 + e^{x/y})dx + (1 - x/y)e^{x/y}dy = 0$

Example-14 Solve the equation: $(x^3 + 3xy^2)dx + (y^3 + 3x^2y)dy = 0$

Example-15 Solve the equation: $(x^2 + y^2)dy + 2x(x + y)dx = 0$

Example-16 Solve the equation: $x^2 \frac{dy}{dx} + xy = \frac{y^3}{x}$, when $y(1) = 1$.

Solution Given differential equation,

$$\begin{aligned} &x^2 \frac{dy}{dx} + xy = \frac{y^3}{x} \\ \Rightarrow &x^3 dy + x^2 y dx = y^3 dx \\ \Rightarrow &(x^2 y - y^3)dx + x^3 dy = 0 \end{aligned}$$

Which is a homogeneous differential equation.

So we put $y = vx$, then $dy = v dx + x dv$

Substituting these values in equation (1), we get

$$\begin{aligned} &x^3(v - v^3)dx + x^3(vdx + xdv) = 0 \\ \Rightarrow &(v - v^3)dx + v dx + x dv = 0 \\ \Rightarrow &(2v - v^3)dx + x dv = 0 \\ \Rightarrow &x dv = (v^3 - 2v)dx \\ \Rightarrow &\frac{dv}{v(v^2 - 2)} = \frac{dx}{x} \end{aligned}$$

$$\Rightarrow \frac{1}{2} \left(\frac{v}{v^2 - 2} - \frac{1}{v} \right) dv = \frac{dx}{x}$$

$$\Rightarrow \frac{2v}{v^2 - 2} dv - \frac{2}{v} dv = \frac{4}{x} dx$$

$$\Rightarrow \ln(v^2 - 2) - 2 \ln v = 4 \ln x + \ln c$$

$$\Rightarrow \ln(v^2 - 2) = \ln v^2 + \ln x^4 + \ln c$$

$$\Rightarrow v^2 - 2 = v^2 x^4 c$$

$$\therefore y^2 - 2x^2 = cx^4 y^2$$

which is the general solution of the given differential equation.

To obtain particular solution, we put $x = 1$ and $y = 1$ in equation (2), we get $1 - 2 = c \Rightarrow c = -1$

Putting $c = -1$ in equation (2), we obtain the particular solution

$$\therefore x^4 y^2 - 2x^2 + y^2 = 0 \text{ (Ans.)}$$

Example-17 Solve the equation: $(y + \sqrt{x^2 + y^2})dx - xdy = 0$, when $y(1) = 0$.

Section-2.3

Equations Reducible to Homogeneous Differential Equations

Solved Problems:

Example-1 $\frac{dy}{dx} = \frac{2x+y-2}{3x+y-3}$

Solution Given equation is may be written as

$$(2x+y-2)dx - (3x+y-3)dy = 0 \quad \dots (1)$$

Now we consider the lines

$$2x+y-2=0 \quad \dots (2)$$

and $3x+y-3=0 \quad \dots (3)$

Since the lines (2) and (3) are not parallel, to find the intersection point of (2) and (3), applying cross multiplication, we get

$$\begin{aligned} \frac{x}{-3+2} &= \frac{y}{-6+6} = \frac{1}{2-3} \\ \Rightarrow \frac{x}{-1} &= \frac{y}{0} = \frac{1}{-1} \\ \Rightarrow x &= 1 \text{ and } y = 0 \end{aligned}$$

We put $x=u+1$ and $y=v+0$, then $dx=du$ and $dy=dv$

Substituting these values in equation (1), we obtain

$$(2u+v)du - (3u+v)dv = 0 \quad \dots (4)$$

Equation (4) is a homogeneous differential equation. So, we let $v=uz$, then $dv=udz+zdu$. Putting these values in equation (4), we get

$$\begin{aligned} (2u+uz)du - (3u+uz)(udz+zdu) &= 0 \\ \Rightarrow (2+z)du - (3+z)(udz+zdu) &= 0 \\ \Rightarrow (2+z-3z-z^2)du - (3+z)udz &= 0 \\ \Rightarrow (2-2z-z^2)du &= (3+z)udz \end{aligned}$$

Example-2 Solve : $(4x-y+7)dx = (2x+y-1)dy$

Example-3 Solve $(2x+y+3)dx = (x+2y+3)dy$

Solution Given equation is may be written as

$$(2x+y+3)dx - (x+2y+3)dy = 0 \quad \dots (1)$$

Now we consider the lines

$$2x+y+3=0 \quad \dots (2)$$

and $x+2y+3=0 \quad \dots (3)$

Since the lines (2) and (3) are not parallel, to find the

$$\Rightarrow \frac{du}{u} - \frac{(3+z)}{(2-2z-z^2)} dz = 0$$

$$\Rightarrow \frac{du}{u} + \frac{(z+3)dz}{z^2+2z-2} = 0$$

$$\Rightarrow \frac{du}{u} + \frac{\left\{ \frac{1}{2}(2z+2)+2 \right\} dz}{z^2+2z-2} = 0$$

$$\Rightarrow \frac{du}{u} + \frac{1}{2} \int \frac{(2z+2)dz}{z^2+2z-2} + 2 \int \frac{dz}{z^2+2z-2} = 0$$

$$\Rightarrow \frac{du}{u} + \frac{1}{2} \int \frac{(2z+2)dz}{z^2+2z-2} + 2 \int \frac{dz}{(z+1)^2 - (\sqrt{3})^2} = 0$$

$$\Rightarrow \ln u + \frac{1}{2} \ln(z^2+2z-2) + \frac{2}{2\sqrt{3}} \ln \left(\frac{z+1-\sqrt{3}}{z+1+\sqrt{3}} \right) = \frac{1}{2} \ln c$$

$$\Rightarrow \ln u^2 + \ln(z^2+2z-2) + \frac{2}{\sqrt{3}} \ln \left(\frac{z+1-\sqrt{3}}{z+1+\sqrt{3}} \right) = c$$

$$\Rightarrow \ln \left(\frac{v^2}{u^2} + \frac{2v}{u} - 2 \right) u^2 + \frac{2}{\sqrt{3}} \ln \left(\frac{v/u+1-\sqrt{3}}{v/u+1+\sqrt{3}} \right) = c$$

$$\Rightarrow \ln(v^2+2uv-2u^2) + \frac{2}{\sqrt{3}} \ln \left\{ \frac{v+(1-\sqrt{3})u}{v+(1+\sqrt{3})u} \right\} = c$$

Since $x=u+1$ and $y=v+0$, so putting $u=x-1$ and $v=y$ in the above equation, we obtain the general solution

$$\ln \left\{ y^2 + 2(x-1)y - 2(x-1)^2 \right\} + \frac{2}{\sqrt{3}} \ln \left\{ \frac{y+(1-\sqrt{3})(x-1)}{y+(1+\sqrt{3})(x-1)} \right\} = c$$

intersection point of (2) and (3), applying cross multiplication, we get

$$\frac{x}{3-6} = \frac{y}{3-6} = \frac{1}{4-1}$$

$$\Rightarrow x = -1 \text{ and } y = -1$$

We put $x=u-1$ and $y=v-1$, then $dx=du$ and $dy=dv$

Substituting these values in equation (1), we obtain

$$(2u + v)du - (u + 2v)dv = 0 \quad \dots (4)$$

Equation (4) is a homogeneous differential equation. So, we let $v = uz$, then $dv = u dz + z du$. Putting these values in equation (4), we get

$$\begin{aligned} (2u + uz)du - (u + 2uz)(udz + zdu) &= 0 \\ \Rightarrow (2 + z)du - (1 + 2z)(udz + zdu) &= 0 \\ \Rightarrow (2 + z - z - 2z^2)du - (1 + 2z)udz &= 0 \\ \Rightarrow (2 - 2z^2)du &= (1 + 2z)udz \\ \Rightarrow \frac{du}{u} + \frac{(2z + 1)}{2(z^2 - 1)}dz &= 0 \\ \Rightarrow \int \frac{du}{u} + \frac{1}{2} \int \frac{2z + 1}{z^2 - 1}dz + \frac{1}{2} \int \frac{dv}{z^2 - 1} &= 0 \end{aligned}$$

Example-4 Solve $(2x - 6y + 3)dx = (x - 3y - 1)dy$

Example-5 Solve : $\frac{dy}{dx} = \frac{6x - 2y - 7}{3x - y + 4}$

Solution Given differential equation is

$$\frac{dy}{dx} = \frac{2(3x - y) - 7}{3x - y + 4} \quad \dots (1)$$

We put $3x - y = z$ then $3 - \frac{dy}{dx} = \frac{dz}{dx}$ or, $\frac{dy}{dx} = 3 - \frac{dz}{dx} \dots (2)$

Putting these values in equation (1), we get

$$\begin{aligned} 3 - \frac{dz}{dx} &= \frac{2z - 7}{z + 4} \\ \Rightarrow 3 - \frac{2z - 7}{z + 4} &= \frac{dz}{dx} \\ \Rightarrow \frac{dz}{dx} &= \frac{3z + 12 - 2z + 7}{z + 4} \end{aligned}$$

Example-6 $(2x - y + 3)dy = (2x - y - 5)dx$

Example-7 $(2x + y + 1)dx + (4x + 2y - 1)dy = 0$

$$\Rightarrow \ln u + \frac{1}{2} \ln(z^2 - 1) + \frac{1}{2} \cdot \frac{1}{2 \cdot 1} \ln \left(\frac{z - 1}{z + 1} \right) = \frac{1}{4} \ln c$$

$$\Rightarrow \ln \left\{ u^4 \cdot (z + 1)^2 (z - 1)^2 \left(\frac{z - 1}{z + 1} \right) \right\} = \ln c$$

$$\Rightarrow u^4 (z + 1)(z - 1)^3 = c$$

$$\Rightarrow u^4 \left(\frac{v}{u} + 1 \right) \left(\frac{v}{u} - 1 \right)^3 = c \left[\because z = \frac{v}{u} \right]$$

$$\Rightarrow (u + v)(v - u)^3 = c$$

Since $x = u - 1$ and $y = v - 1$, so putting $u = x + 1$ and $v = y + 1$ in the above equation, we obtain the general solution:

$$\begin{aligned} (x + 1 + y + 1)(y + 1 - x - 1)^3 &= c \\ \Rightarrow (x + y + 2)(y - x)^3 &= c \quad \text{(Ans.)} \end{aligned}$$

$$\Rightarrow \frac{dz}{dx} = \frac{z + 19}{z + 4}$$

$$\Rightarrow \frac{(z + 4)}{(z + 19)} dz = dx$$

$$\Rightarrow \frac{\{(z + 19) - 15\} dz}{z + 19} = dx$$

$$\Rightarrow \left(1 - \frac{15}{z + 19} \right) dz = dx$$

Integrating both sides, we get

$$z - 15 \ln(z + 19) = x + c$$

$$\Rightarrow 3x - y - 15 \ln(3x - y + 19) = x + c \quad [\because z = 3x - y]$$

$$\therefore 2x - y - 15 \ln(3x - y + 19) = c \quad \text{(Ans.)}$$

Section-2.4

Exact Differential Equations

2.4.1 Exact Differential

Let us consider a differential expression $M(x, y)dx + N(x, y)dy$. If the expression can be written in the differential form $dF(x, y)$, then the expression $M(x, y)dx + N(x, y)dy$ is called exact differential.

Example: The differential expression $y dx + x dy$ can be written in the differential form $d(xy)$. So, the differential expression $y dx + x dy$ is exact differential. And we can write it as $y dx + x dy = d(xy)$.

2.4.2 Exact Differential Equation

A first-order differential equation of the form

$$M(x, y)dx + N(x, y)dy = 0 \quad \dots (1)$$

is said to be an **exact differential equation** if the expression on the left-hand side is an exact differential.

In case of exact differential equation, if there is any function $F(x, y)$ can be found that has for its total differential the expression $M(x, y)dx + N(x, y)dy$; that is

$$dF(x, y) = M(x, y)dx + N(x, y)dy \quad \dots (2)$$

Then certainly, $F(x, y) = c \quad \dots (3)$

is a set of solutions of (1).

Because, from (3), it follows that

$$dF(x, y) = 0$$

From calculus, we obtain

$$\frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy = 0$$

If we put $\frac{\partial F}{\partial x} = M(x, y)$ and $\frac{\partial F}{\partial y} = N(x, y)$, then we obtain

$$M(x, y)dx + N(x, y)dy = 0 \text{ as desired.}$$

Example: The differential equation $(x + y)dx + (x - y)dy = 0$ is exact, because the expression on left hand side can be written as exact differential $d\left\{\frac{1}{2}(x^2 + 2xy - y^2)\right\}$.

And its solution will be $\frac{1}{2}(x^2 + 2xy - y^2) = \frac{c}{2}$ or, $x^2 + 2xy - y^2 = c$

Or, A first-order differential equation of the form

$$M(x, y)dx + N(x, y)dy = 0$$

is said to be an exact differential equation if the condition

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \text{ is satisfied.}$$

Example: The differential equation

$$(6x + y^2)dx + y(2x - 3y)dy = 0$$

is exact. Because, if we let $M = 6x + y^2$ and $N = y(2x - 3y) = 2xy - 3y^2$ then we get, $\frac{\partial M}{\partial y} = 2y$ and $\frac{\partial N}{\partial x} = 2y$. That is

the desired condition $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ is satisfied.

Theorem 2.4.1: If $M, N, \frac{\partial M}{\partial y}$ and $\frac{\partial N}{\partial x}$ are continuous functions of x and y , then a necessary and sufficient condition that $M(x, y)dx + N(x, y)dy = 0$ be an exact equation is that $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.

Proof : Try yourself.

Solved Problems:

Example-1 Solve the differential equation $3x(xy - 2)dx + (x^3 + 2y)dy = 0$.

Solution Given differential equation

$$3x(xy - 2)dx + (x^3 + 2y)dy = 0 \quad \dots (1)$$

Now, comparing the equation (1) with the equation $M(x, y)dx + N(x, y)dy = 0$, we get

$$\begin{aligned} M &= 3x(xy - 2) = 3x^2y - 6x & \text{and} & \quad N = x^3 + 2y \\ \therefore \frac{\partial M}{\partial y} &= 3x^2 & \text{and} & \quad \frac{\partial N}{\partial x} = 3x^2 \end{aligned}$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, we conclude that the differential equation is exact. And, the solution of the given differential equation is $F(x, y) = c$, where

$$\frac{\partial F}{\partial x} = M = 3x^2y - 6x \quad \dots (2)$$

$$\text{and } \frac{\partial F}{\partial y} = N = x^3 + 2y \quad \dots (3)$$

Integrating on both sides of (2) with respect to x , holding y constant, we get

$$F = x^3y - 3x^2 + T(y) \quad \dots (4)$$

where $T(y)$ is arbitrary constant.

Differentiate (4) partially with respect to y , we get

$\begin{aligned} \frac{\partial F}{\partial y} &= x^3 + T'(y) \\ \Rightarrow x^3 + 2y &= x^3 + T'(y) \\ \Rightarrow T'(y) &= 2y \end{aligned}$	$\begin{aligned} &\Rightarrow \frac{d}{dy}\{T(y)\} = 2y \\ &\Rightarrow d\{T(y)\} = 2y dy \\ &\Rightarrow T(y) = y^2 \end{aligned}$	here no arbitrary constant needed, since the solution contain one arbitrary constant c. Thus the solution of the given differential equation is $x^3y - 3x^2 + y^2 = c$ (Ans.)
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Alternative Method: (Method of grouping)

Since, the given differential equation is exact, rearranging the terms of the equation (1), it may be written as

$\begin{aligned} 3x^2y dx - 6x dx + x^3 dy + 2y dy &= 0 \\ \Rightarrow (3x^2y dx + x^3 dy) - 6x dx + 2y dy &= 0 \end{aligned}$	$\begin{aligned} &\Rightarrow d(x^3y) - d(3x^2) + d(y^2) = 0 \\ &\Rightarrow d(x^3y - 3x^2 + y^2) = 0 \end{aligned}$
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On integrating, we thus obtain the solution of the given differential equation is $x^3y - 3x^2 + y^2 = c$. **(Ans.)**

Example-2 Solve $3x(xy - 2)dx + (x^3 + 2y)dy = 0$

Solution Given equation is

$$3x(xy - 2)dx + (x^3 + 2y)dy = 0 \quad \dots (1)$$

Now, comparing the equation (1) with the equation $M(x, y)dx + N(x, y)dy = 0$, we get

$$M = 3xy - 6x \text{ and } N = x^3 + 2y$$

$$\frac{\partial M}{\partial y} = 3x^2 \quad \text{and} \quad \frac{\partial N}{\partial x} = 3x^2$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ so equation (1) is exact.

In this case, the general solution of the given exact differential equation is obtain from the formula given by

$$\begin{aligned} & \int_{y \text{ constant}} M dx + \int (\text{terms of } N \text{ not constaining } x) dy = C \\ \Rightarrow & \int_{y \text{ constant}} (3x^2y - 6x) dx + \int 2y dy = C \\ \Rightarrow & y \int 3x^2 dx - \int 6x dx + \int 2y dy = C \\ \therefore & x^3y - 3x^2 + y^2 = C \text{ (Ans.)} \end{aligned}$$

Example-3 Solve : $(1 + e^{x/y})dx + e^{x/y}\left(1 - \frac{x}{y}\right)dy = 0$

Solution Given differential equation is

$$(1 + e^{x/y})dx + e^{x/y}\left(1 - \frac{x}{y}\right)dy = 0 \quad \dots (1)$$

Now, comparing the equation (1) with the equation $M(x, y)dx + N(x, y)dy = 0$, we get

$$\begin{aligned} M &= 1 + e^{x/y} \text{ and } N = e^{x/y}\left(1 - \frac{x}{y}\right) \\ \therefore \frac{\partial M}{\partial y} &= e^{x/y}\left(\frac{-x}{y^2}\right) \Rightarrow \frac{\partial M}{\partial y} = -\frac{xe^{x/y}}{y^2} \\ \text{and } \frac{\partial N}{\partial x} &= e^{x/y}\left(0 - \frac{1}{y}\right) + \left(1 - \frac{x}{y}\right)e^{x/y} \cdot \frac{1}{y} \\ \Rightarrow \frac{\partial N}{\partial x} &= e^{x/y}\left(-\frac{1}{y} + \frac{1}{y} - \frac{x}{y^2}\right) \end{aligned}$$

$$\Rightarrow \frac{\partial N}{\partial x} = \frac{-xe^{x/y}}{y^2}$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, so the equation (1) is exact.

In this case, the general solution of the given exact differential equation is obtain from the formula given by

$$\begin{aligned} & \int_{y \text{ constant}} M dx + \int (\text{terms of } N \text{ not constaining } x) dy = C \\ \Rightarrow & \int_{y \text{ constant}} (1 + e^{x/y}) dx + \int (0) dy = C \\ \Rightarrow & x + \frac{e^{x/y}}{1/y} = C \\ \therefore & x + ye^{x/y} = C \text{ (Ans.)} \end{aligned}$$

Section-2.5

Special Types of Integrating Factor

2.5.1. Integrating Factor

Definition: If the differential equation

$$M(x, y)dx + N(x, y)dy = 0 \quad \dots (1)$$

is not exact in a particular domain, but after multiplying the equation by a factor $\mu(x, y)$ the reduce equation

$$\mu(x, y) \cdot M(x, y)dx + \mu(x, y) \cdot N(x, y)dy = 0 \quad \dots (2)$$

becomes exact in that domain, then the factor $\mu(x, y)$ is called integrating factor.

Theorem-2.5.1: Suppose the differential equation of the form $M(x, y)dx + N(x, y)dy = 0$ is not exact. If

$\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = f(x)$, a function of x only, then $e^{\int f(x)dx}$ is an integrating factor of the differential equation.

Theorem-2.5.2: Suppose the differential equation of the form $M(x, y)dx + N(x, y)dy = 0$ is not exact. If

$\frac{1}{M} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = g(y)$, a function of y only, then $e^{-\int g(y)dy}$ is an integrating factor of the differential equation.

Theorem-2.5.3: If M and N are homogeneous functions in x, y of the same degree, and if $Mx + Ny \neq 0$, then

$\frac{1}{Mx + Ny}$ is an integrating factor of the differential equation $M(x, y)dx + N(x, y)dy = 0$.

Theorem-2.5.4: If the differential equation $M(x, y)dx + N(x, y)dy = 0$ can be written in the form

$yf(xy)dx + xg(xy)dy = 0$, which is not exact, but $Mx - Ny \neq 0$, then $\frac{1}{Mx - Ny}$ is an integrating factor of the differential equation.

Solved Problems:

Example-1 Solve the Differential Equation $(x^2 + y^2 + x)dx + xydy = 0$

Solution Given Differential Equation

$$(x^2 + y^2 + x)dx + xydy = 0 \quad \dots (1)$$

Now, comparing the equation (1) with the equation $M(x, y)dx + N(x, y)dy = 0$, we get

$$\begin{aligned} M &= x^2 + y^2 + x & \text{and} & & N &= xy \\ \therefore \frac{\partial M}{\partial y} &= 2y & \text{and} & & \frac{\partial N}{\partial x} &= y \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, we conclude that the differential equation is not exact.

Now, we find $\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = 2y - y = y$

$$\Rightarrow \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{y}{xy} = \frac{1}{x} = f(x)$$

Hence, integrating factor $I.F. = e^{\int f(x)dx} = e^{\int \frac{1}{x}dx} = e^{\ln x} = x$

Multiplying on both sides of equation (1) by the $I.F. = x$, we get

$$(x^3 + xy^2 + x^2)dx + x^2y dy = 0 \quad \dots (2)$$

which is, now, an exact differential equation. And, the solution of the given differential equation is $F(x, y) = c$, where

$$\frac{\partial F}{\partial x} = x^3 + xy^2 + x^2 \quad \dots (3)$$

$$\text{and } \frac{\partial F}{\partial y} = x^2y \quad \dots (4)$$

Integrating on both sides of (3) with respect to x , holding y constant, we get

$$F = \frac{x^4}{4} + \frac{x^2y^2}{2} + \frac{x^3}{3} + T(y) \quad \dots (5)$$

where $T(y)$ is arbitrary constant.

Differentiate (5) partially with respect to y , we get

$$\begin{aligned} \frac{\partial F}{\partial y} &= x^2y + T'(y) \\ \Rightarrow x^2y &= x^2y + T'(y) \\ \Rightarrow T'(y) &= 0 \\ \Rightarrow d\{T(y)\} &= 0 \\ \Rightarrow T(y) &= 0 \end{aligned}$$

here no arbitrary constant needed, since the solution contain one arbitrary constant c .

Thus the solution of the given differential equation is $\frac{x^4}{4} + \frac{x^2y^2}{2} + \frac{x^3}{3} = c$ **(Ans.)**

Alternative Method: (Method of grouping)

Since, the given differential equation is exact, rearranging the terms of the equation (2), it may be written as

$$\begin{aligned} x^3dx + xy^2dx + x^2dx + x^2y dy &= 0 \\ \Rightarrow x^3dx + x^2dx + \{xy^2dx + x^2y dy\} &= 0 \\ \Rightarrow d\left(\frac{x^4}{4}\right) + d\left(\frac{x^3}{3}\right) + d\left(\frac{x^2y^2}{3}\right) &= 0 \\ \Rightarrow d\left(\frac{x^4}{4} + \frac{x^3}{3} + \frac{x^2y^2}{3}\right) &= 0 \end{aligned}$$

On integrating, we thus obtain the solution of the given differential equation is $\frac{x^4}{4} + \frac{x^3}{3} + \frac{x^2y^2}{3} = c$ **(Ans.)**

Example-2 Solve: $[y^2(x+1) + y]dx + (2xy+1)dy = 0$

Solution Given differential equation is

$$[y^2(x+1) + y]dx + (2xy+1)dy = 0 \quad \dots (1)$$

Now, comparing the equation (1) with the equation $M(x, y)dx + N(x, y)dy = 0$, we get

$$\begin{aligned} M &= xy^2 + y^2 + y \text{ and } N = 2xy + 1 \\ \therefore \frac{\partial M}{\partial y} &= 2xy + 2y + 1 \text{ and } \frac{\partial N}{\partial x} = 2y \end{aligned}$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, so the equation (1) is not exact.

$$\text{Now, } \frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = 2xy + 2y + 1 - 2y$$

$$\Rightarrow \frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = 2xy + 1$$

$$\therefore \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{2xy+1}{2xy+1} = 1 = f(x)$$

$$\therefore I.F. = e^{\int f(x)dx} = e^{\int dx} = e^x$$

Multiplying both sides of (1) by e^x , we get

$$(xy^2 + y^2 + y)e^x dx + (2xy + 1)e^x dy = 0 \quad \dots (2)$$

Equation (2) is now exact,

Here, $M' = (xy^2 + y^2 + y)e^x$ and $N' = (2xy + 1)e^x$

$$\therefore \int M dx + \int N dy = c$$

$$\Rightarrow \int_{y \text{ constant}} (xy^2 + y^2 + y)e^x dx + \int (0) dy = c$$

$$\Rightarrow y^2 \int x e^x dx + y^2 \int e^x dx + y \int e^x dx = 0$$

$$\Rightarrow y^2 \left[x \int e^x dx - \int 1 \cdot e^x dx \right] + y^2 e^x + y e^x = c$$

$$\Rightarrow y^2 [x e^x - e^x] + y^2 e^x + y e^x = c$$

$$\Rightarrow (xy^2 - y^2 + y^2 + y)e^x = c$$

$$\therefore (xy^2 + y e^x) = c \quad \textbf{(Ans.)}$$

Example-3 Solve : $(12y + 4y^3 + 6x^2)dx + 3(x + xy^2)dy = 0$

Solution Given differential equation is

$$(12y + 4y^3 + 6x^2)dx + 3(x + xy^2)dy = 0 \quad \dots (1)$$

Now, comparing the equation (1) with the equation $M(x,y)dx + N(x,y)dy = 0$, we get

$$M = 12y + 4y^3 + 6x^2 \text{ and } N = 3x(1 + y^2)$$

$$\therefore \frac{\partial M}{\partial y} = 12 + 12y^2 \text{ and } \frac{\partial N}{\partial x} = 3(1 + y^2)$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, so the equation (1) is not exact.

$$\text{Now, } \frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = (12 + 12y^2) - 3(1 + y^2)$$

$$\Rightarrow \frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = 9(1 + y^2)$$

$$\therefore \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{9(1 + y^2)}{3x(1 + y^2)} = \frac{3}{x} = f(x)$$

$$\therefore \text{I.F.} = e^{\int \frac{3}{x} dx} = e^{3 \ln x} = e^{\ln x^3} = x^3$$

Multiplying both sides of (1) by x^3 , we get

$$\begin{aligned} (12y + 4y^3 + 6x^2)x^3 dx + 3(x + xy^2)x^3 dy &= 0 \\ \Rightarrow (12x^3y + 4x^3y^3 + 6x^5)dx + 3(x^4 + x^4y^2)dy &= 0 \end{aligned} \quad \dots (2)$$

Now equation (1) is exact.

$$\text{Here } M' = 12x^3y + 4x^3y^3 + 6x^5 \text{ and } N' = 3(x^4 + x^4y^2)$$

$$\begin{aligned} \therefore \int_{y \text{ constant}} M dx + \int (\text{terms of } N \text{ not containing } x) dy &= c \\ \Rightarrow \int_{y \text{ constant}} (12x^3y + 4x^3y^3 + 6x^5)dx + \int (0)dy &= c \\ \Rightarrow 12y \int x^3 dx + 4y^3 \int x^3 dx + 6 \int x^5 dx &= c \\ \Rightarrow 12y \frac{x^4}{4} + 4y^3 \cdot \frac{x^4}{4} + 6 \frac{x^6}{6} &= c \\ \therefore 3x^4y + x^4y^3 + x^6 &= c \quad (\text{Ans.}) \end{aligned}$$

Example-4 Find integrating factor and solve $\left(y + \frac{1}{3}y^3 + \frac{1}{2}x^2\right)dx + \frac{1}{4}(x + xy^2)dy = 0$.

Solution Given differential equation is

$$\left(y + \frac{1}{3}y^3 + \frac{1}{2}x^2\right)dx + \frac{1}{4}(x + xy^2)dy = 0 \quad \dots (1)$$

Now, comparing the equation (1) with the equation $M(x, y)dx + N(x, y)dy = 0$, we get

$$\begin{aligned} M &= y + \frac{1}{3}y^3 + \frac{1}{2}x^2 \text{ and } N = \frac{1}{4}(x + xy^2) \\ \therefore \frac{\partial M}{\partial y} &= 1 + y^2 \text{ and } \frac{\partial N}{\partial x} = \frac{1}{4}(1 + y^2) \\ \text{Now, } \frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} &= (1 + y^2) - \frac{1}{4}(1 + y^2) = \frac{3}{4}(1 + y^2) \\ \therefore \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) &= \frac{\frac{3}{4}(1 + y^2)}{\frac{1}{4}(x + xy^2)} = \frac{3}{x} = f(x) \end{aligned}$$

$$\therefore \text{I.F.} = e^{\int f(x) dx} = e^{\int \frac{3}{x} dx} = e^{3 \ln x} = e^{\ln x^3} = x^3$$

Multiplying both sides of (1) by x^3 , we get

$$\begin{aligned} \therefore \left(x^3y + \frac{1}{3}x^3y^3 + \frac{1}{2}x^5\right)dx + \frac{1}{4}(1 + y^2)x^4dy &= 0 \\ x^3y dx + \frac{1}{3}x^3y^3 dx + \frac{1}{2}x^5 dx + \frac{1}{4}x^4 dy + \frac{1}{4}x^4 y^2 dy &= 0 \\ \Rightarrow \left(x^3y dx + \frac{1}{4}x^4 dy\right) + \left(\frac{1}{3}x^3y^3 dx + \frac{1}{4}x^4 y^2 dy\right) + \frac{1}{2}x^5 dx &= 0 \end{aligned}$$

$$\Rightarrow d\left(\frac{x^4 y}{4}\right) + d\left(\frac{x^4 y^3}{12}\right) + d\left(\frac{x^6}{12}\right) = 0$$

Integrating this both sides, we get

$$\Rightarrow \frac{x^4 y}{4} + \frac{1}{12} x^4 y^3 + \frac{1}{12} x^6 = \frac{c}{12}$$

$$\therefore 3x^4 y + x^4 y^3 + x^6 = C \quad (\text{Ans.})$$

Example-5 Find an integrating factor to solve $(4xy + 3y^2 - x)dx + x(x + 2y)dy = 0$.

Solution Given differential equation is

$$(4xy + 3y^2 - x)dx + x(x + 2y)dy = 0 \quad \dots (1)$$

Now, comparing the equation (1) with the equation $M(x, y)dx + N(x, y)dy = 0$, we get

$$M = 4xy + 3y^2 - x \text{ and } N = x(x + 2y) = x^2 + 2xy$$

$$\therefore \frac{\partial M}{\partial y} = 4x + 6y \text{ and } \frac{\partial N}{\partial x} = 2x + 2y$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, so the equation (1) is not exact.

$$\text{Now, } \frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = 4x + 6y - 2x - 2y = 2x + 4y = 2(x + 2y)$$

$$\therefore \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{2(x + 2y)}{x(x + 2y)} = \frac{2}{x} = f(x)$$

$$\therefore \text{I.F.} = e^{\int \frac{2}{x} dx} = e^{2 \ln x} = e^{\ln x^2} = x^2 \quad (\text{Ans.})$$

Now multiplying both sides of (1) by x^2 , we get

$$(4x^3 y + 3x^2 y^2 - x^3)dx + (x^4 + 2x^3 y)dy = 0$$

Which is now exact equation. Rearranging the terms of (2), we get

$$\begin{aligned} 4x^3 y dx + 3x^2 y^2 dx - x^3 dx + x^4 dy + 2x^3 y dy &= 0 \\ \Rightarrow (4x^3 y dx + x^4 dy) + (3x^2 y^2 dx + 2x^3 y dy) - x^3 dx &= 0 \\ d(x^4 y) + d(x^3 y^2) - d\left(\frac{x^4}{4}\right) &= 0 \\ \Rightarrow d\left(x^4 y + x^3 y^2 - \frac{x^4}{4}\right) &= 0 \\ \Rightarrow x^4 y + x^3 y^2 - \frac{x^4}{4} &= \frac{c}{4} \\ \therefore 4x^4 y + 4x^3 y^2 - x^4 &= c \quad (\text{Ans.}) \end{aligned}$$

Example-6 Solve : $(1 + xy)y dx + (1 - xy)x dy = 0$

Solution Given differential equation is

$$(1 + xy)y dx + (1 - xy)x dy = 0 \quad \dots (1)$$

Now, comparing the equation (1) with the equation $M(x, y)dx + N(x, y)dy = 0$, we get

$$M = (1 + xy)y \text{ and } N = (1 - xy)x$$

$$\text{or, } M = y + xy^2 \text{ and } N = x - x^2 y$$

$$\therefore \frac{\partial M}{\partial y} = 1 + 2xy \text{ and } \frac{\partial N}{\partial x} = 1 - 2xy$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, so the equation (1) is not exact. But the equation is of the form $f(xy)y dx + g(xy)x dy = 0$.

$$\text{Here, } Mx - Ny = (y + xy^2)x - (x - x^2 y)y$$

$$\Rightarrow Mx - Ny = xy + x^2y^2 - xy + x^2y^2$$

$$\Rightarrow Mx - Ny = 2x^2y^2 \neq 0$$

$\therefore$ The integrating factor

$$\text{I.F.} = \frac{1}{Mx - Ny} = \frac{1}{2x^2y^2}$$

Now, multiplying on both sides of (1) by $\frac{1}{2x^2y^2}$, we get

$$\frac{(1+xy)y}{2x^2y^2}dx + \frac{(1-xy)x}{2x^2y^2}dy = 0$$

$$\Rightarrow \left(\frac{1+xy}{2x^2y} \right)dx + \left(\frac{1-xy}{2xy^2} \right)dy = 0$$

$$\Rightarrow \left(\frac{1}{x^2y} + \frac{1}{x} \right)dx + \left(\frac{1}{xy^2} - \frac{1}{y} \right)dy = 0$$

which is an exact differential equation

Now applying grouping method, we obtain

$$\left(\frac{1}{x^2y}dx + \frac{1}{xy^2}dy \right) + \frac{dx}{x} - \frac{dy}{y} = 0$$

$$\Rightarrow d\left(-\frac{1}{xy} \right) + \frac{dx}{x} - \frac{dy}{y} = 0$$

$$\Rightarrow -\frac{1}{xy} + \ln x - \ln y = c$$

$$\therefore \ln \frac{x}{y} - \frac{1}{xy} = c \text{ (Ans.)}$$

Alternative Method :

$$\text{Here } M = \frac{1}{x^2y} + \frac{1}{x} \text{ and } N = \frac{1}{xy^2} - \frac{1}{y}$$

The general solution of the given differential equation can be obtained from the formula,

$$\therefore \int_{y \text{ constant}} M dx + \int (\text{terms of } N \text{ not containing } x) dy = c$$

$$\Rightarrow \int_{y \text{ constant}} \left(\frac{1}{x^2y} + \frac{1}{x} \right) dx + \int \left(-\frac{1}{y} \right) dy = c$$

$$\Rightarrow \frac{1}{y} \int \frac{dx}{x^2} + \int \frac{dx}{x} - \int \frac{dy}{y} = c$$

$$\Rightarrow \frac{1}{y} \left(-\frac{1}{x} \right) + \ln x - \ln y = c$$

$$\therefore \ln \frac{x}{y} - \frac{1}{xy} = c \text{ (Ans.)}$$

Section-2.6

Integrating factor by Inspection

2.6.1 Method of Inspection

In this section, we are concerned with equations that are simple enough to enable us to find integrating factors by inspection. The ability to do this depends largely upon recognition of certain common exact differentials and upon experience.

2.6.2 List of Some Important Exact Differentials

Below are five exact differentials that occur frequently:

	Differential expression	Integrating factor	Exact differential
1.	$x dy + y dx$	1	$x dy + y dx = d(xy)$
2.	$x dy - y dx$	$\frac{1}{x^2}$	$\frac{x dy - y dx}{x^2} = d\left(\frac{y}{x}\right)$
3.	$y dx - x dy$	$\frac{1}{y^2}$	$\frac{y dx - x dy}{y^2} = d\left(\frac{x}{y}\right)$
4.	$x dy - y dx$	$\frac{1}{x^2 + y^2}$	$\frac{x dy - y dx}{x^2 + y^2} = d\left(\tan^{-1} \frac{y}{x}\right)$
5.	$x dy - y dx$	$\frac{1}{x^2 - y^2}$	$\frac{x dy - y dx}{x^2 - y^2} = d\left(\frac{1}{2} \ln \frac{x+y}{x-y}\right)$

There are some more exact differentials which we should remember to solve in this method easily:

	Differential expression	Integrating factor	Exact differential
1.	$x dy - y dx$	$\frac{1}{xy}$	$\frac{dy}{y} - \frac{dx}{x} = d(\ln y - \ln x) = d\left(\ln \frac{y}{x}\right)$
2.	$x dy - y dx$	$\frac{1}{(x-y)^2}$	$\frac{x dy - y dx}{x^2 - y^2} = d\left[\frac{1}{2} \left(\frac{x+y}{x-y}\right)\right]$
3.	$x dy - y dx$	$\frac{1}{x\sqrt{x^2 - y^2}}$	$\frac{x dy - y dx}{x\sqrt{x^2 - y^2}} = d\left[\sin^{-1} \frac{y}{x}\right]$
4.	$x dy - y dx$	$\frac{1}{(x+y)^2}$	$-\frac{y dx - x dy}{(x+y)^2} = d\left[-\frac{1}{2} \left(\frac{x-y}{x+y}\right)\right]$
5.	$x dy + y dx$	$\frac{1}{xy}$	$\frac{dx}{x} + \frac{dy}{y} = d[\ln(x+y)]$
6.	$x dy + y dx$	$\frac{1}{x^2 y^2}$	$\frac{x dy + y dx}{x^2 y^2} = d\left(-\frac{1}{xy}\right)$
7.	$x dy + y dx$	$\frac{1}{x^2 + y^2}$	$\frac{x dy + y dx}{x^2 + y^2} = d\left[\frac{1}{2} \ln(x^2 + y^2)\right]$
8.	$dx + dy$	$\frac{1}{x+y}$	$\frac{dx + dy}{x+y} = d[\ln(x+y)]$

Solved Problems:

Example-1 Show that, $\frac{1}{x^2} f\left(\frac{y}{x}\right)$, is an integrating factor of the differential equation $x dy - y dx = 0$.

Solution Given differential equation is

$$x dy - y dx = 0 \quad \dots(1)$$

Multiplying both sides of equation (1) by $\frac{1}{x^2} f\left(\frac{y}{x}\right)$, we obtain $\frac{f\left(\frac{y}{x}\right)}{x^2}$

$$\begin{aligned} \frac{1}{x} f\left(\frac{y}{x}\right) dy - \frac{y}{x^2} f\left(\frac{y}{x}\right) dx &= 0 \\ \Rightarrow \frac{y}{x^2} f\left(\frac{y}{x}\right) dx - \frac{1}{x} f\left(\frac{y}{x}\right) dy &= 0 \end{aligned} \quad \dots(2)$$

Now, comparing the equation (1) with the equation $M(x,y)dx + N(x,y)dy = 0$, we get

$$M = \frac{y}{x^2} f\left(\frac{y}{x}\right) \quad \text{and} \quad N = -\frac{1}{x} f\left(\frac{y}{x}\right)$$

Then, $\frac{\partial M}{\partial y} = \frac{1}{x^2} f\left(\frac{y}{x}\right) + \frac{y}{x^2} f'\left(\frac{y}{x}\right)\left(\frac{1}{x}\right) = \frac{1}{x^2} f\left(\frac{y}{x}\right) + \frac{y}{x^3} f'\left(\frac{y}{x}\right)$

and $\frac{\partial N}{\partial x} = \frac{1}{x^2} f\left(\frac{y}{x}\right) - \frac{1}{x} f'\left(\frac{y}{x}\right)\left(\frac{-y}{x^2}\right) = \frac{1}{x^2} f\left(\frac{y}{x}\right) + \frac{y}{x^3} f'\left(\frac{y}{x}\right)$

We observe that, $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$. Hence $\frac{1}{x^2} f\left(\frac{y}{x}\right)$ is an integrating factor of the given differential equation. **[Shown]**

Example-2 Solve the differential equation $y dx + (x + x^3 y^2) dy = 0$

Solution Given differential equation

$$y dx + (x + x^3 y^2) dy = 0 \quad \dots(1)$$

Let us group the terms of like degree, we write the equation as

$$\begin{aligned} (y dx + x dy) + x^3 y^2 dy &= 0 \\ \Rightarrow d(xy) + x^3 y^2 dy &= 0 \quad [\because y dx + x dy = d(xy)] \end{aligned}$$

Dividing both sides by $x^3 y^3$, we obtain the equation

$$\begin{aligned} \Rightarrow \frac{d(xy)}{(xy)^3} + \frac{dy}{y} &= 0 \\ \Rightarrow \int \frac{d(xy)}{(xy)^3} + \int \frac{dy}{y} &= 0 \quad [\text{Integrating both sides}] \\ \Rightarrow -\frac{1}{2x^2 y^2} + \ln y &= -\ln c \\ \Rightarrow \ln(cy) &= \frac{1}{2x^2 y^2} \\ \therefore 2x^2 y^2 \ln(cy) &= 1 \quad (\text{Ans.}) \end{aligned}$$

Example-3 Solve : $x dx + y dy + (x^2 + y^2) dy = 0$

Solution Given differential equation is

$$\begin{aligned} & x dx + y dy + (x^2 + y^2) dy = 0 \\ \Rightarrow & \frac{x dx + y dy}{x^2 + y^2} + dy = 0 & [\text{Dividing both sides by } x^2 + y^2] \\ \Rightarrow & \frac{1}{2} \cdot \frac{d(x^2 + y^2)}{x^2 + y^2} + dy = 0 \end{aligned}$$

Integrating on both sides, we get

$$\begin{aligned} & \frac{1}{2} \ln(x^2 + y^2) + y = \frac{c}{2} \\ \therefore & \ln(x^2 + y^2) + 2y = c \end{aligned}$$

which is the required solution.

Example-4 Solve : $x dy - y dx = \sqrt{x^2 + y^2} dx$

Solution Given differential equation,

$$x dy - y dx = \sqrt{x^2 + y^2} dx$$

Dividing both sides by x^2 , we get

$$\begin{aligned} & \frac{x dy - y dx}{x^2} = \frac{\sqrt{x^2 + y^2}}{x} \cdot \frac{dx}{x} \\ \Rightarrow & d\left(\frac{y}{x}\right) = \sqrt{1 + \left(\frac{y}{x}\right)^2} \cdot \frac{dx}{x} & \left[\because \frac{x dy - y dx}{x^2} = d\left(\frac{y}{x}\right) \right] \\ \Rightarrow & \frac{d(y/x)}{\sqrt{1 + (y/x)^2}} = \frac{dx}{x} \\ \Rightarrow & \frac{dz}{\sqrt{1 + z^2}} = \frac{dx}{x} & \left[\because z = \frac{y}{x} \right] \end{aligned}$$

Integrating this we get

$$\begin{aligned} & \ln\left(z + \sqrt{1 + z^2}\right) = \ln x + \ln c & \left[\because \int \frac{dz}{\sqrt{1 + z^2}} = \ln(z + \sqrt{1 + z^2}) \right] \\ \Rightarrow & \ln\left(\frac{y}{x} + \sqrt{1 + \left(\frac{y}{x}\right)^2}\right) = \ln cx \\ \Rightarrow & \frac{y}{x} + \frac{\sqrt{x^2 + y^2}}{x} = cx \\ \therefore & y + \sqrt{x^2 + y^2} = cx^2 \text{ (Ans.)} \end{aligned}$$

Example-5 $3x^2 y dx + (y^4 - x^3) dy = 0$

Solution Given differential equation,

$$3x^2y dx + (y^4 - x^3) dy = 0 \quad \dots(1)$$

Two terms in the coefficients of dx and dy are of degree three, and the other coefficient is not of degree three. Let us regroup the terms to get

$$\Rightarrow (3x^2y dx - x^3 dy) + y^4 dy = 0$$

$$\Rightarrow [y d(x^3) - x^3 dy] + y^4 dy = 0$$

$$\Rightarrow \frac{y d(x^3) - x^3 dy}{y^2} + y^2 dy = 0 \quad [\text{Dividing both sides by } y^2]$$

$$\Rightarrow d\left(\frac{x^3}{y}\right) + y^2 dy = 0 \quad \left[\because \frac{vdu - u dv}{v^2} = d\left(\frac{u}{v}\right) \right]$$

$$\Rightarrow \frac{x^3}{y} + \frac{y^3}{3} = \frac{C}{3}$$

$$\therefore 3x^3 + y^4 = C \text{ (Ans.)}$$

Example-6 Solve the differential equation

$$x \cos \frac{y}{x} (y dx + x dy) = y \sin \frac{y}{x} (x dy - y dx)$$

Solution Given differential equation

$$x \cos \frac{y}{x} (y dx + x dy) = y \sin \frac{y}{x} (x dy - y dx) \quad \dots(1)$$

Since we know the exact differentials $d(xy) = y dx + x dy$ and $\frac{x dy - y dx}{x^2} = d\left(\frac{y}{x}\right)$, the equation (1) write in this way

$$x \cdot d(xy) = y \tan \frac{y}{x} \cdot \frac{x dy - y dx}{x^2} \cdot x^2$$

$$\Rightarrow \frac{d(xy)}{xy} = \tan\left(\frac{y}{x}\right) \cdot d\left(\frac{y}{x}\right) \quad [\text{Dividing both sides by } xy]$$

Integrating on both sides, we get

$$\Rightarrow \ln(xy) = \ln \sec\left(\frac{y}{x}\right) + \ln C$$

$$\therefore xy = C \cdot \sec\left(\frac{y}{x}\right) \text{ (Ans.)}$$

Example-7 Solve $\frac{dy}{dx} = \frac{2x - y + 1}{x + 2y - 3}$

Solution Given differential equation,

$$\frac{dy}{dx} = \frac{2x - y + 1}{x + 2y - 3}$$

$$\Rightarrow (2x - y + 1)dx = (x + 2y - 3)dy$$

$$\Rightarrow 2x dx - y dx + dx - x dy - 2y dy + 3dy = 0$$

$$\Rightarrow 2x dx - 2y dy - (x dy + y dx) + dx + d(3y) = 0$$

$$\Rightarrow d(x^2) - d(y^2) - d(xy) + dx + d(3y) = 0$$

$$\Rightarrow d(x^2 - y^2 - xy + x + 3y) = 0$$

Integrating on both sides, we obtain the required

$$\text{solution } x^2 - y^2 - xy + x + 3y = c \text{ (Ans.)}$$

Section-2.7

Linear Differential Equations

2.7.1. Linear Differential Equations

Definition: An equation is said to be linear and of order one in the dependent variable y and independent variable x , if it is written in the form

$$\frac{dy}{dx} + P(x)y = Q(x) \quad \dots (1)$$

Or, The differential equation of the form $\frac{dy}{dx} + P(x)y = Q(x)$ is called first order linear differential equation.

Example: The differential equation, $2(y - 4x^2)dx + xdy = 0$ is a first order linear differential equation, because it can be written in the standard form: $\frac{dy}{dx} + \frac{2}{x}y = 8x$, where $x \neq 0$; which is of the form (1), where $P(x) = \frac{2}{x}$ and $Q(x) = 8x$.

Theorem-2.7.1: The linear differential equation of order one

$$\frac{dy}{dx} + P(x)y = Q(x) \quad \dots(1)$$

has an integrating factor of the form $e^{\int P(x)dx}$.

Proof: The differential equation (1) can be written as

$$\{P(x)y - Q(x)\}dx + dy = 0 \quad \dots(2)$$

The differential equation (2) is of the form

$$M(x, y)dx + N(x, y)dy = 0$$

where, $M(x, y) = P(x)y - Q(x)$ and $N(x, y) = 1$

$$\therefore \frac{\partial M}{\partial y} = P(x) \text{ and } \frac{\partial N}{\partial x} = 0$$

So, the differential equation (1) will be exact, if $P(x) = 0$. In this case, the differential equation reduces to a simple separable equation.

Let us consider, $P(x) \neq 0$. In this case the equation (2) possesses an integrating factor that depends on x only and let the integrating factor be $I.F. = \mu(x)$.

Now, multiplying equation (2) by $\mu(x)$, we get

$$\{\mu(x)P(x)y - \mu(x)Q(x)\}dx + \mu(x)dy = 0 \quad \dots (3)$$

By definition, $\mu(x)$ is an integrating factor of equation (3) if and only if equation (3) is exact, i.e., if it satisfy the condition

$$\frac{\partial}{\partial y}[\mu(x)P(x)y - \mu(x)Q(x)] = \frac{\partial}{\partial x}[\mu(x)]$$

$$\text{or, } \mu(x)P(x) = \frac{d}{dx}[\mu(x)] \quad \dots (4)$$

In differential equation (4), $P(x)$ is a known function and $\mu(x)$ is an unknown function, which we are trying to determine. The differential equation (4) is separable; separating the variables, we have $\frac{d\mu}{\mu} = P(x)dx$.

Integrating, we obtain the particular solution $\ln|\mu| = \int P(x)dx$ or, $\mu = e^{\int P(x)dx}$.

Note: (i) The linear differential equation of order one $\frac{dy}{dx} + P(x)y = Q(x)$ has an integrating factor of the form $e^{\int P(x)dx}$.
(ii) The linear differential equation of order one $\frac{dx}{dy} + P(y)x = Q(y)$ has an integrating factor of the form $e^{\int P(y)dy}$.

Theorem-2.7.2: A one-parameter family of solutions of the equation

$$\frac{dy}{dx} + P(x)y = Q(x) \quad \dots (1)$$

$$\text{is } y e^{\int P(x)dx} = \int e^{\int P(x)dx} \cdot Q(x)dx + C$$

$$\text{or } y = e^{-\int P(x)dx} \left[\int e^{\int P(x)dx} \cdot Q(x)dx + C \right]$$

Proof: Since the equation (1) is of first order linear differential equation, its integrating factor is of the form $\mu(x) = e^{\int P(x)dx}$.

Now, multiplying both sides of the equation (1) by $e^{\int P(x)dx}$, we get

$$e^{\int P(x)dx} \cdot \frac{dy}{dx} + e^{\int P(x)dx} \cdot P(x)y = e^{\int P(x)dx} Q(x)$$

$$\text{or, } e^{\int P(x)dx} dy + e^{\int P(x)dx} P(x)y dx = e^{\int P(x)dx} Q(x)dx$$

$$\text{or, } d\left(e^{\int P(x)dx} \cdot y\right) = e^{\int P(x)dx} Q(x)dx$$

Integrating this we obtain the solution of the equation (1) in the form

$$e^{\int P(x)dx} \cdot y = \int e^{\int P(x)dx} Q(x)dx + C$$

$$\text{or, } y = e^{-\int P(x)dx} \cdot \int e^{\int P(x)dx} Q(x)dx + C \text{ [Proved]} \quad \dots (2)$$

Note: One parameter family of solutions (2) of the differential equation (1) includes all solutions of the differential equation (1).

Note:

(i) The general solution of the linear differential equation

$$\frac{dx}{dy} + xP(y) = Q(y) \text{ is } x e^{\int P(y)dy} = \int Q(y) e^{\int P(y)dy} + C.$$

(ii) If, $Q(y) = 0$ then $\frac{dx}{dy} + P(y)x = 0$ and its general solution is $x = c e^{-\int P(y)dy}$.

Solved Problems:

Example-1 Solve the Differential Equation: $(1+x^2)\frac{dy}{dx} + y = \tan^{-1}x$.

Solution Given Differential Equation $(1+x^2)\frac{dy}{dx} + y = \tan^{-1}x \quad \dots (1)$

Can be written in the standard form $\frac{dy}{dx} + \frac{y}{1+x^2} = \frac{\tan^{-1}x}{1+x^2}$.

which is of the form $\frac{dy}{dx} + P(x)y = Q(x)$.

where, $P(x) = \frac{1}{1+x^2}$ and $Q(x) = \frac{\tan^{-1} x}{1+x^2}$

So, its integrating factor $I.F. = e^{\int \frac{dx}{1+x^2}} = e^{\tan^{-1} x}$.

Now, multiplying both sides of equation (2) by $e^{\tan^{-1} x}$, we get

$$e^{\tan^{-1} x} dy + e^{\tan^{-1} x} \cdot \frac{y}{1+x^2} dx = \frac{e^{\tan^{-1} x} \cdot \tan^{-1} x}{1+x^2} dx$$

$$\text{or, } d(ye^{\tan^{-1} x}) = \frac{e^{\tan^{-1} x} \cdot \tan^{-1} x}{1+x^2} dx$$

Integrating on both sides we obtain

$$ye^{\tan^{-1} x} = \int \frac{e^{\tan^{-1} x} \cdot \tan^{-1} x}{1+x^2} dx \dots (2)$$

For right hand side of (2), we put

$$\tan^{-1} x = u \text{ and } \frac{1}{1+x^2} dx = du$$

Then equation (2) becomes

$$ye^{\tan^{-1} x} = \int ue^u du$$

$$\text{or, } ye^{\tan^{-1} x} = u \int e^u du - \int \left\{ \frac{du}{du} \int e^u du \right\} du \text{ [integrating by parts]}$$

$$\text{or, } ye^{\tan^{-1} x} = ue^u - \int e^u du$$

$$\text{or, } ye^{\tan^{-1} x} = ue^u - e^u + C$$

$$\text{or, } ye^{\tan^{-1} x} = e^u (u - 1) + C$$

$$\text{or, } ye^{\tan^{-1} x} = e^{\tan^{-1} x} (\tan^{-1} x - 1) + C$$

$$\therefore y = (\tan^{-1} x - 1) + Ce^{-\tan^{-1} x}$$

which is the desired solution of the given differential equation.

Example-2 Solve the differential equation: $(1+x^2) \frac{dy}{dx} + 2xy = 4x^2$

Solution Given differential equation is

$$(1+x^2) \frac{dy}{dx} + 2xy = 4x^2$$

$$\Rightarrow \frac{dy}{dx} + \frac{2x}{1+x^2} y = \frac{4x^2}{1+x^2} \dots (1)$$

which is of the standard form $\frac{dy}{dx} + P(x)y = Q(x)$,

where $P(x) = \frac{2x}{1+x^2}$ and $Q(x) = \frac{4x^2}{1+x^2}$.

$\therefore$ Integrating factor of (1) is

$$I.F. = e^{\int \frac{2x}{1+x^2} dx} = e^{\ln(1+x^2)} = 1+x^2$$

Example-3 Solve the differential equation: $\cos^2 x \frac{dy}{dx} + y = \tan x$.

Solution Given differential equation is

$$\cos^2 x \frac{dy}{dx} + y = \tan x$$

$$\Rightarrow \frac{dy}{dx} + y \sec^2 x = \sec^2 x \tan x \dots (1)$$

which is of the standard form $\frac{dy}{dx} + P(x)y = Q(x)$, where $P(x) = \sec^2 x$ and $Q(x) = \sec^2 x \tan x$

$\therefore$ Integrating factor of (1) is $I.F. = e^{\int P(x) dx} = e^{\int \sec^2 x dx} = e^{\tan x}$.

Now multiplying both sides of equation (1) by $e^{\tan x}$, we get

$$e^{\tan x} \frac{dy}{dx} + ye^{\tan x} \cdot \sec^2 x = \sec^2 x \cdot \tan x \cdot e^{\tan x}$$

$$\Rightarrow \frac{d}{dx}(ye^{\tan x}) = \sec^2 x \cdot \tan x \cdot e^{\tan x}$$

Now multiplying both sides of equation (1) by

$$1+x^2, \text{ we get: } (1+x^2) \frac{dy}{dx} + 2xy = 4x^2$$

$$\Rightarrow (1+x^2) dy + 2xy dx = 4x^2 dx$$

$$\Rightarrow d\{y(1+x^2)\} = 4x^2 dx$$

$$\Rightarrow \int d\{y(1+x^2)\} = \int 4x^2 dx$$

$$\Rightarrow y(1+x^2) = \frac{4}{3} x^3 + \frac{c}{3}$$

$$\therefore 3y(1+x^2) = 4x^3 + c \text{ (Ans.)}$$

$$\Rightarrow ye^{\tan x} = \int \sec^2 x \tan x e^{\tan x} dx + c \quad [\text{Integrating on both sides}]$$

$$= \int \tan x e^{\tan x} d(\tan x) + c = \tan x e^{\tan x} - e^{\tan x} + c$$

$$\Rightarrow y = e^{-\tan x} (\tan x e^{\tan x} - e^{\tan x}) + ce^{-\tan x} = \tan x - 1 + ce^{-\tan x}$$

which is the required solution.

Example-4 Solve the differential equation : $\frac{dy}{dx} + y \tan x - \sec x = 0$

Example-5 Solve the differential equation: $\frac{dy}{dx} + \frac{y}{x} = x^2$

Example-6 Solve the differential equation: $(1-x^2)\frac{dy}{dx} - xy = 1$

Example-7 Solve the differential equation: $x\frac{dy}{dx} - 2y = (x-2)e^x$

Example-8 Solve the Differential Equation $ydx + (xy^2 + x - y)dy = 0$

Solution Given Differential Equation

$$ydx + (xy^2 + x - y)dy = 0 \quad \dots (1)$$

can be re-written as

$$\frac{dx}{dy} + \frac{xy^2 + x - y}{y} = 0$$

$$\text{or, } \frac{dx}{dy} + \frac{xy^2 + x - y}{y} = 0$$

$$\text{or, } \frac{dx}{dy} + \frac{y^2 + 1}{y}x - 1 = 0$$

$$\text{or, } \frac{dx}{dy} + \frac{y^2 + 1}{y}x = 1 \quad \dots (2)$$

which is standard form of a linear differential equation of order one. So, its integrating factor

$$I.F. = e^{\int \frac{y^2+1}{y} dy} = e^{\int \left(y + \frac{1}{y}\right) dy} = e^{\frac{y^2}{2} + \ln y} = e^{\frac{y^2}{2}} \cdot e^{\ln y} = ye^{\frac{y^2}{2}}$$

Now, multiplying on both sides of equation (2) by

$\frac{y^2}{ye^{\frac{y^2}{2}}}$, we get

$$\frac{y^2}{ye^{\frac{y^2}{2}}} \cdot \frac{dx}{dy} + \frac{y^2}{ye^{\frac{y^2}{2}}} \cdot \frac{y^2+1}{y}x = \frac{y^2}{ye^{\frac{y^2}{2}}}$$

$$\text{or, } \frac{y^2}{ye^{\frac{y^2}{2}}} dx + \frac{y^2}{ye^{\frac{y^2}{2}}} \cdot \frac{y^2+1}{y}x dy = \frac{y^2}{ye^{\frac{y^2}{2}}} dy$$

$$\text{or, } d\left(xy e^{\frac{y^2}{2}}\right) = d\left(e^{\frac{y^2}{2}}\right)$$

Integrating on both sides we obtain the desired solution

$$xy e^{\frac{y^2}{2}} = e^{\frac{y^2}{2}} + C$$

$$\text{or, } xy = 1 + Ce^{-\frac{y^2}{2}} \quad (\text{Ans.})$$

Example-9 Solve the differential equation: $(x+y+1)\frac{dy}{dx} = 1$

Example-10 Solve the differential equation: $(1+y^2)dx = (\tan^{-1}y - x)dy$.

Section 2.8

Bernoulli's Differential Equation

2.8.1. Bernoulli's Differential Equation

Definition: The differential equation of the form: $\frac{dy}{dx} + P(x)y = Q(x)y^n$, is called a Bernoulli's differential equation.

We observe that, if $n=0$, the equation reduces to a first order linear differential equation. And if $n=1$, then the linear equation is a separable equation.

Example: The differential equation: $\frac{dy}{dx} + y = xy^3$ is called a Bernoulli's differential equation, where $P(x)=1$, $Q(x)=x$ and $n=3$.

2.8.2. Solution of Bernoulli's Differential Equation

Theorem-2.5.3: Let us consider $n \neq 0, 1$. Then the transformation $v = y^{1-n}$ reduces the Bernoulli's differential equation to a linear equation in v .

Proof: Bernoulli's differential equation is

$$\frac{dy}{dx} + P(x)y = Q(x)y^n \quad \dots (1)$$

Dividing both sides by y^n , we get

$$y^{-n} \cdot \frac{dy}{dx} + y^{1-n} \cdot P(x) = Q(x) \quad \dots (2)$$

If we put $v = y^{1-n}$ and $\frac{dv}{dx} = (1-n)y^{-n} \frac{dy}{dx}$ or,

$y^{-n} \frac{dy}{dx} = \frac{1}{(1-n)} \frac{dv}{dx}$, then equation (2) reduces to

$$\frac{1}{(1-n)} \frac{dv}{dx} + P(x)v = Q(x)$$

$$\text{or } \frac{dv}{dx} + (n-1)P(x)v = (n-1)Q(x)$$

Letting $P_1(x) = (n-1)P(x)$ and $Q_1(x) = (n-1)Q(x)$, then the above equation becomes

$$\frac{dv}{dx} + P_1(x)v = Q_1(x)$$

which is a first order linear differential equation in v .

Theorem-2.7.4: A one-parameter family of solutions of the Bernoulli's differential equation:

$$\frac{dy}{dx} + P(x)y = Q(x)y^n \quad \dots (1)$$

$$\text{is } v e^{\int (n-1)P(x)dx} = (n-1) \int e^{\int (n-1)P(x)dx} \cdot Q(x)dx + C \quad \dots (2)$$

Solved Problems:

Example-1 Solve the Differential Equation $\frac{dy}{dx} + y = y^2 e^x$.

Solution Given Differential Equation

$$\frac{dy}{dx} + y = y^2 e^x$$

is a Bernoulli's differential equation. Dividing both sides by y^2 , we get

$$\frac{1}{y^2} \frac{dy}{dx} + \frac{1}{y} = e^x \quad \dots (1)$$

We put $\frac{1}{y} = z$, then $-\frac{1}{y^2} \frac{dy}{dx} = \frac{dz}{dx}$ or, $\frac{1}{y^2} \frac{dy}{dx} = -\frac{dz}{dx}$

Putting these values in equation (1), we get

$$-\frac{dz}{dx} + z = e^x$$

$$\text{or, } \frac{dz}{dx} - z = -e^x \quad \dots (2)$$

integrating factor

$$I.F. = e^{\int (-1)dx} = e^{-x}$$

Now, multiplying both sides of (2) by e^{-x} , we get

$$e^{-x} \frac{dz}{dx} - ze^{-x} = -e^{-x} \cdot e^x$$

$$\Rightarrow e^{-x} dz - ze^{-x} dx = -1$$

$$\Rightarrow d(ze^{-x}) = -1$$

$$\Rightarrow ze^{-x} = -x + c \quad [\text{Integrating}]$$

$$\Rightarrow \frac{1}{y} = (c-x)e^x \quad [\because z = \frac{1}{y}]$$

$$\therefore (c-x)ye^x = 1$$

which is the desired solution of the given differential equation.

Example-2 Solve : $\frac{dy}{dx} + y = y^2 e^x$.

Solution Given equation is

$$\frac{dy}{dx} + y = y^2 e^x$$

is a Bernoulli's differential equation. Dividing both sides by y^2 , we get

$$\frac{1}{y^2} \frac{dy}{dx} + \frac{1}{y} = e^x \quad \dots (1)$$

We put $\frac{1}{y} = z$ then $\frac{1}{y^2} \frac{dy}{dx} = -\frac{dz}{dx}$... (2)

Putting these values in equation (1), we get

$$\begin{aligned} -\frac{dz}{dx} + z &= e^x \\ \Rightarrow \frac{dz}{dx} - z &= -e^x \quad \dots (3) \end{aligned}$$

which is a first order linear differential equation and

Example-3 Solve : $\frac{dy}{dx} + \frac{y}{x} = \frac{y^2}{x^2}$

Example-4 Solve: $\frac{dy}{dx} + \frac{1}{x} \sin 2y = x^3 \cos^2 y$

Solution Given differential equation is

$$\frac{dy}{dx} + \frac{1}{x} \sin 2y = x^3 \cos^2 y$$

is a Bernoulli's differential equation. Dividing both sides by $\cos^2 y$, we get

$$\begin{aligned} \frac{1}{\cos^2 y} \frac{dy}{dx} + \frac{1}{x} \frac{\sin y \cos y}{\cos^2 y} &= x^3 \\ \Rightarrow \sec^2 y \frac{dy}{dx} + \frac{2 \tan y}{x} &= x^3 \quad \dots (1) \end{aligned}$$

We put $\tan y = z$ then $\sec^2 y \frac{dy}{dx} = \frac{dz}{dx}$

Putting these values in equation (1), we get

$$\frac{dz}{dx} + \frac{2z}{x} = x^3 \quad \dots (2)$$

which is a first order linear differential equation and

its integrating factor

$$\text{I.F.} = e^{-\int 1 dx} = e^{-x}$$

Multiplying both sides of (2) by e^{-x} we get

$$\begin{aligned} \frac{d}{dx} [ze^{-x}] &= -e^x \cdot e^{-x} \\ \Rightarrow \frac{d}{dx} [ze^{-x}] &= -1 \end{aligned}$$

Integrating both sides w. r. to x , we get

$$\begin{aligned} ze^{-x} &= -x + c \\ \Rightarrow \frac{1}{y} \cdot e^{-x} &= -x + c \\ \Rightarrow \frac{1}{ye^x} + x &= c \\ \Rightarrow 1 + xye^x &= cye^x \quad (\text{Ans.}) \end{aligned}$$

its integrating factor

$$\text{I.F.} = e^{\int \frac{2}{x} dx} = e^{2 \ln x} = e^{\ln x^2} = x^2$$

Multiplying both sides of (2) by x^2 , we get

$$\begin{aligned} x^2 \frac{dz}{dx} + 2zx &= x^5 \\ \Rightarrow x^2 dz + 2zx \, dx &= x^5 dx \\ \Rightarrow d(zx^2) &= x^5 dx \end{aligned}$$

Integrating both sides w. r. to x , we get

$$\begin{aligned} zx^2 &= \frac{x^6}{6} + \frac{c}{6} \\ \Rightarrow x^2 \tan y &= \frac{x^6}{6} + \frac{c}{6} \\ \therefore 6x^2 \tan y &= x^6 + c \quad (\text{Ans.}) \end{aligned}$$

Section-2.9

Riccati's Equation

2.9.1 Riccati's Equation

The equation

$$\frac{dy}{dx} = A(x)y^2 + B(x)y + C(x) \quad \dots (1)$$

is called (or known as) Riccati's differential equation.

Theorem-2.9.1: Show that if $A(x)=0$ for all x , then equation (1) is a linear differential equation, whereas if $C(x)=0$ for all x , then equation (1) is a Bernoulli's differential equation.

Proof: Try yourself.

Theorem-2.9.2: Suppose that some particular solution y_1 of Riccati's equation is known, then a more general solution containing one arbitrary constant can be obtained through the substitution:

$$y = y_1(x) + \frac{1}{v(x)} \quad \dots (2)$$

Then reduces (1) to a first order linear equation by $v(x)$. That is, $v(x)$ satisfies the first order linear equation

$$\frac{dv}{dx} = -(B + 2Ay_1)v - A \quad \dots (3)$$

Proof: Try yourself.

Solved Problems:

Example-1 Find a one-parameter family of solutions of the Riccati's equation:

$$\frac{dy}{dx} = (1-x)y^2 + (2x-1)y - x \text{ given particular solution } y_1(x)=1.$$

Solution Given differential equation

$$\frac{dy}{dx} = (1-x)y^2 + (2x-1)y - x \quad \dots (1)$$

Also given particular solution $y_1(x)=1$.

Now, we use substitution $y = y_1(x) + \frac{1}{v(x)}$

$$\text{i.e., } y = 1 + \frac{1}{v} \text{ or, } \frac{dy}{dx} = -\frac{1}{v^2} \frac{dv}{dx}$$

Then equation (1) becomes,

$$\begin{aligned} -\frac{1}{v^2} \frac{dv}{dx} &= (1-x) \left(1 + \frac{1}{v}\right)^2 + (2x-1) \left(1 + \frac{1}{v}\right) - x \\ \Rightarrow -\frac{dv}{dx} &= (1-x)(v^2 + 2v + 1) + (2x-1)(v + v^2) - xv^2 \\ \Rightarrow -\frac{dv}{dx} &= v^2(1-x+2x-1-x) + v(2-2x+2x-1) + (1-x) \\ \Rightarrow -\frac{dv}{dx} &= v + (1-x) \\ \therefore \frac{dv}{dx} + v &= x-1 \quad \dots (2) \end{aligned}$$

Equation (2) is a first order linear differential equation.

Integrating factor of equation (2) is, I.F. = $e^{\int(1)dx} = e^x$

Now multiplying equation (2) by e^x , we obtain

$$e^x \frac{dv}{dx} + e^x v = e^x(x-1)$$

$$\begin{aligned}
&\Rightarrow e^x dv + e^x v dx = e^x (x-1) dx \\
&\Rightarrow d(ve^x) = e^x (x-1) dx \\
&\Rightarrow ve^x = (x-1) \int e^x dx - \int \left\{ \frac{d}{dx} (x-1) \int e^x dx \right\} dx \text{ [Integrating on both sides]} \\
&\Rightarrow ve^x = (x-1)e^x - e^x + c \quad [\text{where } c \text{ is an arbitrary constant}] \\
&\Rightarrow \frac{e^x}{y-1} = (x-2)e^x + c \\
&\therefore y = 1 + \frac{1}{(x-2) + e^{-x}c}
\end{aligned}$$

Which is the required one-parameter family of solutions of the given Riccati's equation.

Exercise:

In exercise 01-05, Verify that the given function is a particular solution of the given Riccati equation. Then use appropriate transformation to solve the following Riccati equations:

- $\frac{dy}{dx} = -y^2 + xy + 1$; given particular solution $y_1(x) = x$.
- $\frac{dy}{dx} = -8xy^2 + 4x(4x+1)y - (8x^3 + 4x^2 - 1)$; given particular solution $y_1(x) = x$. **[Ans. $y = (2 + ce^{-2x^2})^{-1} + x$]**
- $\frac{dy}{dx} = y^2 - 2xy + x^2 + 1$; given particular solution $y_1(x) = x$. **[Ans. $y = (c - x)^{-1} + x$]**
- $\frac{dy}{dx} = y^2 - \frac{y}{x} - \frac{1}{x^2}$; given particular solution $y_1(x) = \frac{1}{x}$. **[Ans. $y = 2x(c - x^2)^{-1} + x^{-1}$]**
- $\frac{dy}{dx} = \frac{2\cos^2 x - \sin^2 x + y^2}{2\cos x}$; given particular solution $y_1(x) = \sin x$. **[Ans. $y = (c \cos x - \frac{1}{2} \sin x)^{-1} + \sin x$]**
- The propagation of a single action in a large population (for example, drivers turning on headlights at sunset) often depends partly on external circumstances (gathering darkness) and partly on a tendency to imitate others who have already performed the action in question. In this case the proportion $y(t)$ of people who have performed the action can be described by the equation
$$\frac{dy}{dx} = (1-y)[x(t) + by] \quad \dots (A)$$
where $x(t)$ measures the external stimulus and b is the imitation coefficient.
 - Observe that equation (A) is a Riccati equation and that $y_1(t)=1$ is one solution. Use the transformation $y = y_1(t) + \frac{1}{v(t)}$, and find the linear equation satisfied by $v(t)$. **[Ans. $v' - [x(t) + b]v = b$]**
 - Find $v(t)$ in the case that $x(t) = at$, where a is a constant. Leave your answer in the form of an integral. **[Ans. $v = \left[b \int \mu(t) dt + c \right] / \mu(t), \mu(t) = \exp[-(at^2/2) - bt]$]**

Riccati equations are named for Jacopo Francesco Riccati (1676--1754), a Venetian nobleman, who declined university appointments in Italy, Austria, and Russia to pursue his mathematical studies privately at home. Riccati studied these equations extensively. However, it was Euler (in 1760) who discovered the result stated in this problem.